Vijay Sai Chigullapally

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EDUCATION

University of North Texas Denton, TX, USA

M.S. in Computer Science (**GPA:** 4.0/4.0) Expected graduation, May 2026

Course work: Machine Learning, Deep learning, Artificial Intelligence, Database Management System, Big Data, Data Engineering, , Software Engineering.

WORK EXPERIENCE

University of North Texas | Research Assistant - Machine Learning | Denton, TX **Jan 2025 – Present**

- Designed and optimized CNN and Transformer architectures in PyTorch, increasing predictive accuracy by 15% over baseline models through targeted architectural refinements.
- Applied model quantization and pruning techniques to reduce inference latency by 30%, enabling real-time deployment on resource-constrained edge environments.
- Built automated ETL pipelines in Python to process 500GB+ of unstructured research data, improving ingestion throughput by 40% via parallel processing and Parquet serialization.
- Conducted 45+ systematic experiments on hyperparameter sensitivity and architectural bottlenecks, achieving an 18% performance gain over state-of-the-art benchmarks. Documented experimental findings and methodologies for internal publication and peer review.

Kanerika |Data Scientist |Hyderabad, TS, India **January 2024 – June 2024**

- Developed hybrid LSTM and Isolation Forest ensemble models for real-time anomaly detection across 500K+ daily telemetric records, capturing both temporal and spatial outliers.
- Collaborated with the ML team to refine training strategies on 1TB+ of historical logs, improving rare-event recall by 20% using SMOTE and cost-sensitive learning to address 1:1000 class imbalances.
- Implemented signal-to-noise optimization techniques that reduced false-positive rates by 15% across 200+ concurrent sensor streams, significantly reducing alert fatigue for the monitoring team.
- Integrated processed telemetric streams into high-throughput predictive pipelines feeding 10+ automated BI dashboards, enabling data-driven executive decision-making.

Saras Analytics |Data Engineer |Hyderabad, TS, India **May 2023 - October 2023**

- Engineered high-throughput ELT pipelines using Apache Kafka and Saras Daton to consolidate real-time data from 200+ e-commerce sources (Shopify, Amazon, Meta) into Snowflake, reducing ingestion latency by 40%.
- Enhanced Customer 360 models to identify high-value segments and churn triggers, contributing to a 15% uplift in customer retention and a 20% increase in Average Order Value for key retail partners.
- Developed 70+ dbt models and SQL frameworks to transform raw commerce data into structured analytics layers, establishing a unified single source of truth for financial and marketing metrics.
- Deployed automated reporting pipelines surfacing critical KPIs (ROAS, CAC, LTV) for global retail brands, saving 150+ analyst hours monthly and improving cross-channel reporting accuracy by 25%.

TECHNICAL SKILLS

- **Languages:** Python, SQL, R, Java, C++.
- **Machine Learning and AI:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, LangChain, NLP, LLMs, AI Agents, RAG, Model Context Protocol (MCP).
- **Big Data and Data Engineering:** Apache Spark, Apache Kafka, dbt, ETL/ELT Pipelines.
- **Databases:** PostgreSQL, Snowflake, BigQuery, Oracle, Supabase.
- **Cloud and DevOps:** Google Cloud Platform (GCP), Amazon Web Services(AWS), Azure, Docker, Git, GitHub Actions, CI/CD, Linux.
- **Frontend, Backend, APIs:** HTML5, CSS3, Flutter, FastAPI, Flask, REST APIs, Selenium.
- **Data Visualization:** Tableau, Power BI, Matplotlib, Seaborn, Plotly, Excel.

PROJECTS

Fin Bot: AI-Driven Personal Finance Management Tool |*Flutter, FastAPI, Python, Docker, GCP Cloud Run, Supabase, REST APIs, CI/CD* **GitHub**

- Built a cross-platform Flutter and FastAPI chatbot leveraging OCR and ML to automate expense tracking, reducing manual data entry by 70% and improving financial insight accuracy by 20%.
- Generated dynamic spending visualizations (category breakdowns and ttrends) using Matplotlib for personalized financial insights.
- Deployed containerized services on Docker and GCP Cloud Run with secure CI/CD pipelines, achieving 99.5% uptime and reducing API response latency from 400ms to 150ms through asynchronous optimization.

Plant Disease Detection, Classification & Crop Recommender| *Python, TensorFlow, MobileNetV2, Random Forest, Flask, REST APIs* **GitHub**

- Developed a web-based plant phenotyping platform using a CNN (MobileNetV2) trained on the PlantVillage dataset (50K+ images, 38 classes, 14 crops) for disease detection and classification.
- Trained a Random Forest crop recommender (200 trees) on 2,200 samples with 7 agronomic inputs (N, P, K, temperature, humidity, pH, rainfall), achieving ~99.5% accuracy and 99.5% weighted F1 score.
- Delivered end-to-end pipelines from training to Flask-based UI and API inference, streamlining crop decisions and reducing manual diagnosis effort.

Biomedical Image Classifier |*Python, TensorFlow, Keras, ResNet, VGG16, Apache Spark* **GitHub**

- Designed a CNN-based deep learning model using TensorFlow with transfer learning (ResNet) for classification of X-ray and MRI datasets, achieving 90%+ accuracy.
- Leveraged Apache Spark for distributed preprocessing of high-volume medical imaging data, enabling efficient model training on large datasets.

CERTIFICATIONS

Machine Learning with Python: Foundations **Machine Learning**

Model Context Protocol (MCP): Hands-On with Agentic AI **Model Context Protocol**

ETL in python and SQL **ETL (Python and SQL)**